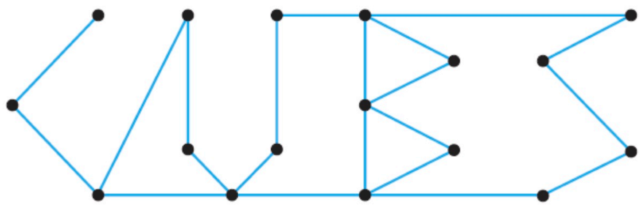
วิรากร ธนพัตร 6809617415

1. CLO 4 (5 คะแนน) Does the following graph have an Euler path? Why or why not?

If you think that this graph has an Euler path, label edges and vertices in the graph properly and state the Euler path you discover.

Hint ในการเขียนแสดง Euler path ที่พบอย่างถูกต้อง นักศึกษาต้องเช็คนิยามของ “path” ในเอกสารประกอบการบรรยาย

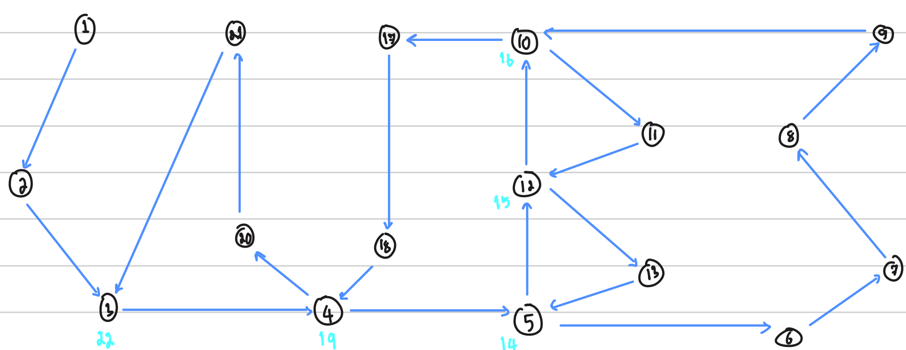


{"Vertex": 17, "Edge": 20}

This graph has only one vertex, which the degree is odd.

So the graph has Euler path.

This is a Euler path which I found. vvv



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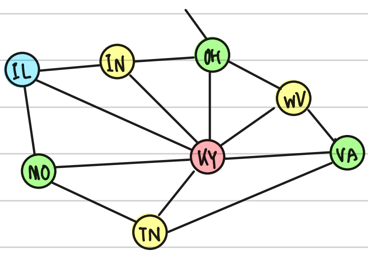
2. CLO 4 (10 คะแนน) Find a map of the United States. Report the map you found. Draw a graph whose vertices represent the states of Illinois, Missouri, Tennessee, Virginia, West Virginia, Ohio, Indiana, and Kentucky. Draw an edge between any two vertices whose states share a common border. Explain why it takes four colors to color this graph (and hence also to distinguish these states on a map).

Note You must provide both a correct definition and the drawing of your graph.

Graph :



To explain why it take four colours in the graph must to know that graph is a planar graph which any edges in the graph not corssing each other.

Then we can paint a color to each vertex by the rule is " we won't paint same color for vertices each share same edge together".

Because of previous purposes so it can write down the graph by taking 4 colours in this graph.

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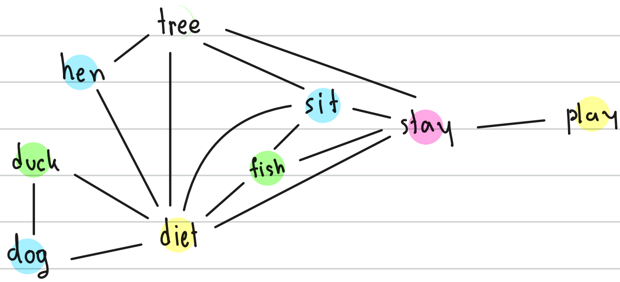
3. CLO 4 (10 คะแนน) Using as few groups as possible, put the words fish, sit, stay, play, diet, tree, duck, dog, and hen into groups such that none of the words in a group have any letters in common.
Use a graph model and graph coloring to solve this problem.

Note You must provide both a correct definition and the drawing of your graph. Also, you must explain how you apply graph coloring to find a solution of this problem.



Group I. diet, play
Group II. tree, fish, duck
Group III. dog, hen, sit
Group IV. stay

To solve this problem we must draw a graph which contain vertices with edges (connect vertices each have same letter(s) in common together) then paint colors for any vertices which not have edge connected each together.

After that the same colour won't have same letter in common for each vertex.

4. CLO 4 (15 คะแนน) วิรากร ธนบัตร 6809617415

A round-robin tournament among four teams — Canadiens, Canucks, Flames, and Oilers — has the following results: Canucks defeat Canadiens; Canucks defeat Flames; Canucks defeat Oilers; Canadiens defeat Oilers; Flames defeat Canadiens; Oilers defeat Flames.

(a) Model these results with a directed graph, where each vertex represents a team and each edge represents a game, pointing from the winner to the loser.

Note You must provide both a correct definition and the drawing of your graph.

(b) Find a circuit in this graph.

(c) Explain why the existence of a circuit in such a graph makes it hard to rank the teams from best to worst.

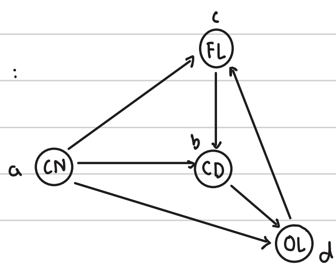
Initial: Canadiens as CD and a

Canuchs as CN and b

Flames as FL and c

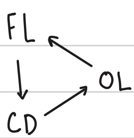
Oilers as OL and d

(a) Graph :



Winner ⟶ loser

(b)



(c) because we can find true winner and loser from circuit by

① - if Flames defeats Canadiens, Flumes would be a winner but not really because Flumes is defeated by Oilers, and same as Flames Oilers is not a winner too because Oilers is defeated by Canadiens {And then loop in ① again}

Therefore we can only know Canucks is 1st but we don't know following sequences. So unpossible to rank them.

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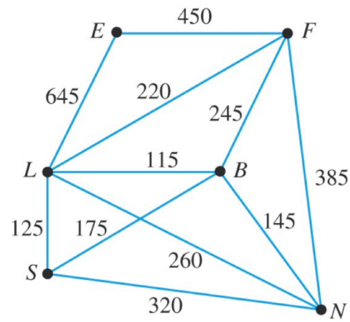
5. CLO 4 (10 **คะแนน**) Consider the network in the following. Plan a round trip starting and ending at San Diego that visits all the other cities in as few miles as possible. In other words, find a circuit that contains every vertex and has minimal weight.

	B	E	F	L	N	S
Barstow						
Eureka						
Fresno	245	450				
Los Angeles	115	645	220			
Needles	145		385	260		
San Diego	175			125	320	



A network showing mileage between cities

$S \to L \to E \to F \to B \to N \to S$ = 1870 miles

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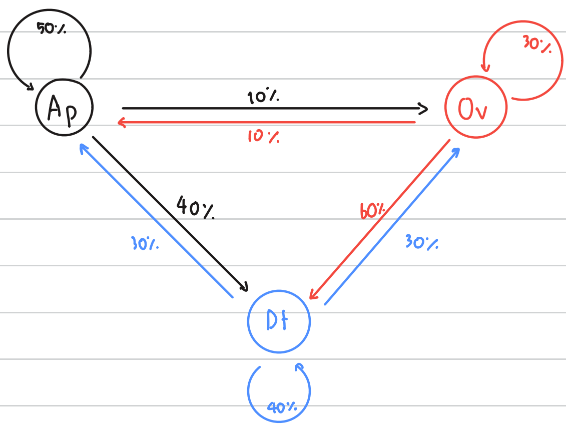
6. CLO 4 (35 คะแนน) Previous Midterm Exam Question

A car rental company has three locations in Mexico City: the International Airport, Oficina Vallejo, and Downtown. Customers can drop off their vehicles at any of these locations. Based on prior experience, the company expects that, at the end of each day, 40% of the cars that begin the day at the Airport will end up Downtown, 50% will return to the Airport, and 10% will be at Oficina Vallejo. Similarly, 60% of the Oficina Vallejo cars will end up Downtown, with 30% returning to Oficina Vallejo and 10% to the Airport. Finally, 30% of the Downtown cars will end up at each of the other locations, with 40% staying at the Downtown location.

Model this situation with a directed network. If the company starts with all of its cars at the Airport, how will the cars be distributed after two days of rentals?

Note You must provide both a correct definition and the drawing of your graph.

Initial Mexico City : The international Airport as Ap and Airport
Oficina Vallejo as Ov
Downtown as Dt



Day I : 50% Ap, 40% Dt, 10% Ov

Day II : 25% + 12% (Dt) + 1% (Ov) Ap, 16% + 20% (Ap) + 6% (Ov) Dt, 3% + 5% (Ap) + 12% (Dt) Ov

↳ 38% in Ap, 42% in Dt, 20% in Ov

Therefore After Two Day, 38% of cars be in Airport, 42% of cars be in Downtown and 20% of cars be in Oficina Vallejo.

วิรากร ธนบัตร 6809617415

7. CLO 4 (15 **คะแนน**) Previous Midterm Exam Question

นักล็อบบี้ทางการเมือง 5 คน ต้องการเข้าพบสมาชิกสภาผู้แทนราษฎรจำนวน 7 ท่าน ได้แก่ Alan, Bill, Carlos, David, Ellen, Francis และ George การเข้าพบจะเป็นลักษณะการประชุมกลุ่มย่อยภายใน 1 วัน โดยนักล็อบบี้ทางการเมืองแต่ละคนต้องการประชุมกับกลุ่มของสมาชิกสภาผู้แทนราษฎรที่แตกต่างกัน ดังนี้

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1. Alan, Bill,และ David
2. Bill, Carlos,และ Francis
3. Alan, Bill, David,และ George
4. Ellen และ George
5. David, Ellen, และ Francis
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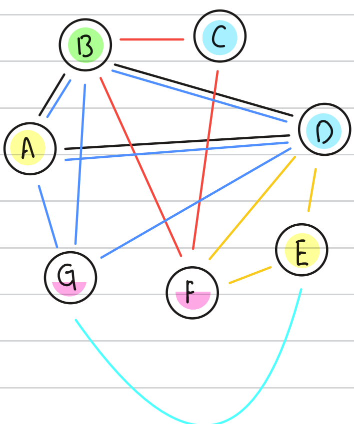
จำนวนสล็อตประชุมที่น้อยที่สุดที่จะต้องจัดขึ้นภายใน 1 วันนี้ เป็นเท่าใด (minimum number of time slots) ? โดยในการจัดสล็อตประชุม มีเงื่อนไขว่าสมาชิกสภาผู้แทนราษฎรทุกคนต้องสามารถเข้าร่วมการประชุมกลุ่มย่อยได้ตามความต้องการของนักล็อบบี้ทางการเมือง

ให้แสดงรูปกราฟ (Graph) อธิบายองค์ประกอบของกราฟ และอภิปรายโดยสังเขปการใช้งานกราฟเพื่อตอบคำถามข้อนี้ด้วย

Initial

Alan as A	Bill as B	Carlos as C	David as D
Ellen as E	Francis as F	George as G	



จากกราฟ edge แต่ละเส้นแสดงถึงความต้องการของนักล็อบบี้ที่ต้องการพบ ส.ส. คนใดบ้าง โดยใช้สีแบ่ง vertex ของนักการเมือง เพื่อดูว่านักการเมืองคนใดสามารถเข้าประชุมในกลุ่มย่อยพร้อมกันได้บ้าง

คำตอบ คือ 4 รอบ